

ISLAVIEW RESORT MANAGEMENT SYSTEM

Complete Capstone Specification Documents & Project Charter Timeline

Project Cycle: 2026 Academic Year **Document Version:** v4.0 (Final Submission)

System State: Phase 3 (Active Implementation)

1. THE PROJECT CHARTER OVERVIEW

As documented in the official project initialization frameworks, the project charter defines the scope boundaries, baseline requirements, and success metrics for the implementation of the **Islaview Resort Management System**.

- **Project Objectives and Constraints:** Deliver a fast, mobile-first booking ecosystem and back-office management console for Islaview Resort by 2026. Hardware integration and local connectivity latencies serve as key operational constraints.
- **Key Stakeholders:**
 - *Guests (Customers):* Seeking frictionless, instantaneous reservation and local e-wallet settlement interfaces.
 - *Admin (Resort Owner/Manager):* Requiring centralized analytics, inventory command CRUD, and dynamic policy logic control.
 - *Staff (Front Desk/Receptionist):* Requiring rapid physical check-in interfaces and live housekeeping toggles.
- **Risks Identified:** Real-time record state desynchronization under high-traffic spikes; transaction timing discrepancies during third-party e-wallet processing; and legacy operational resistance during front-desk system migration.
- **Benefits of the Project:** Elimination of manual book-keeping discrepancies, complete defense against booking overlaps, maximized room asset allocation efficiency, and automatic data compilation for income analysis.
- **General Overview of the Budget:** Structured financial distribution covering environment staging, database resource allocation, automated payment endpoint registration, and personnel system training programs.
- **Success Metrics:** Prompt confirmation latency (<3 seconds), zero over-allocation errors, and seamless user adoption scores over testing cycles.

2. PROJECT IMPLEMENTATION TIMELINE & MILESTONE MATRIX

The following table presents the complete milestone itinerary and workflow distribution as structured in the system roadmap requirements:

PHASE / PERIOD	KEY TARGETS & MILESTONES	DETAILED DELIVERABLES & WORK PACKAGES	STATUS INDICATOR
Phase 1 Feb 02 – Mar 15, 2026	M1: Concept Approval	• Operational problem definition for Islaview Resort. • Drafting core Project Charter parameters (Objectives, Scope, Constraints). • Completing institutional proposal hearing framework presentation.	COMPLETED
Phase 2 Mar 16 – Apr 19, 2026	M2: UI/UX & Schema Freeze	• Mapping Guest mobile layout mockups and paths. • Designing Admin Revenue Dashboard and Inventory interfaces. • Formulating Relational Database Schemas (Rooms, Bookings, Users tables).	COMPLETED
Phase 3 Apr 20 – Jun 14, 2026	M3: Core Logic Integration	[Current Execution Phase] • Developing backend logic and secure active session handling. • Implementing full Inventory management CRUD controls. • Coding the real-time calendar availability matrix and housekeeping status toggles. • Integrating programmatic hooks for GCash, Maya, and PayPal APIs.	ACTIVE PROGRESS
Phase 4 Jun 15 – Jul 19, 2026	M4: System Freeze & Sign-off	• Simulating concurrent guest requests to prevent double-booking anomalies. • Testing webhook endpoints for e-wallet payment status confirmations. • Front-desk usability testing for staff walk-in workflows. • Fixing logic bugs and security testing for guest data.	UPCOMING

PHASE / PERIOD	KEY TARGETS & MILESTONES	DETAILED DELIVERABLES & WORK PACKAGES	STATUS INDICATOR
Phase 5 Jul 20 – Aug 25, 2026	M5: Production Deployment	• Migrating system architecture to live cloud/ local hosting. • Populating production databases with active room inventories. • Handover of analytical dashboards to Islaview Resort administration. • Final capstone defense presentation and project archival.	UPCOMING

3. DETAILED FUNCTIONAL ARCHITECTURE BY USER ROLES

1. GUEST USER MODULE (MOBILE-FIRST INTERFACE)

- **Real-Time Availability:** An interactive calendar visualization providing immediate representation of vacant rooms or cottage structures.
- **Room Gallery & Details:** High-definition imagery accompanied by descriptive lists of structural amenities (Wi-Fi, pool access) and occupancy caps.
- **Booking Engine:** A linear, step-by-step transaction form capturing entry dates, room selections, and occupant profiles cleanly.
- **Secure Payment Portal:** Programmatic transaction gateways providing API links to regional channels (**GCash**, **Maya**, PayPal).
- **Personal Dashboard:** Authenticated client panel to view historical logs and extract electronic reservation vouchers or transaction receipts.
- **Cancellation/Rescheduling Request:** Interactive request submissions triggered via dedicated actions conforming to automated business policies.

2. ADMIN USER MODULE (CENTRAL OPERATIONS MANAGEMENT)

- **Dynamic Dashboard:** Visual statistical summary showing real-time check-ins, check-outs, and accumulated institutional income.
- **Room/Inventory Management:** Unrestricted CRUD (Create, Read, Update, Delete) execution controls for room classification, pricing rules, and descriptions.
- **Seasonal Pricing Logic:** Programmable parameters enabling instant margin adjustment during holidays or peak seasonal operations.
- **Reporting & Analytics:** Statistical extraction tools compiling metrics on popular units and monthly structural income, exportable to PDF/Excel layouts.
- **User Management:** Secure credential generation and Role-Based Access Control (RBAC) distribution for on-site staff.
- **Voucher Verification:** Quick validation utility matching physical check-in documents with secure unique alphanumeric Reference IDs.

3. STAFF USER MODULE (FRONT-DESK OPERATIONS)

- **Check-in/Check-out Module:** Rapid click tracking transforming status metrics from "Confirmed" to "Checked-in" or "Checked-out".
- **Walk-in Booking Interface:** A clean console enabling receptionists to register walk-in guests manually and hold room stock immediately.
- **Housekeeping Status Matrix:** Immediate state toggle allowing staff to mark units as "*Cleaning*", "*Dirty*", or "*Ready for Next Guest*", instantly syncing with the customer-facing calendar.
- **Payment Verification View:** Dedicated validation dashboard allowing front-desk receptionists to visually cross-reference GCash/Maya transaction codes before releasing room access.

4. CORE TECHNICAL MODULES MAPPING

USER GROUP	CORE FEATURE COMPONENT	TECHNICAL IMPLEMENTATION SPEC
GUEST	E-Wallet Payment Portal	API Integration with GCash/Maya endpoints. Secure token handling and automated callback listeners for webhook validation.

USER GROUP	CORE FEATURE COMPONENT	TECHNICAL IMPLEMENTATION SPEC
GUEST	Real-Time Calendar View	AJAX/Fetch async state monitoring mapping room-date arrays to block double-allocation of items.
ADMIN	Seasonal Pricing Logic Engine	Automated database multiplier rules based on specific date ranges, tracking holiday surges.
ADMIN	Reporting & PDF Analytics	Server-side document compilation using modern data export tools to generate operational spreadsheets and logs.
STAFF	Housekeeping Live Status	Immediate database status toggles syncing with real-time UI indicators for front-desk and reservation views.
STAFF	Walk-In Booking Engine	Simplified data-entry form bypassing standard email verification to speed up check-in directly at the counter.